

ARTICLE



Validation of Nanopore long-read sequencing to resolve *RPGR* ORF15 genotypes in individuals with X-linked retinitis pigmentosa

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X-linked retinitis pigmentosa (XLRP) is characterized by progressive vision loss leading to legal blindness in males and a broad severity spectrum in carrier females. Pathogenic alterations of the retinitis pigmentosa GTPase regulator gene (*RPGR*) are responsible for over 70% of XLRP cases. In the retina, the *RPGR*^{ORF15} transcript includes a terminal exon, called ORF15, that is altered in the large majority of *RPGR*-XLRP cases. Unfortunately, due to its highly repetitive sequence, ORF15 represents a considerable challenge in terms of sequencing for molecular diagnostic laboratories. However, in a recent preliminary work Yahya et al. reported a long-read sequencing approach seeming promising. Here, the aim of the study was to validate and integrate this new sequencing strategy in a routine screening workflow. For that purpose, we performed a masked test on 52 genomic DNA samples from male and female individuals carrying 32 different pathogenic ORF15 variations including 20 located in the highly repetitive region of the exon. For the latter, we have obtained a detection rate of 80–85% in males and 60–80% in females after bioinformatic analyses. These numbers raised to 100% for both status after adding a complementary visual inspection of ORF15 long-reads. In accordance with these results, and considering the frequency of ORF15 pathogenic variations in XLRP, we suggest that a long-read screening of ORF15 should be systematically considered before any other sequencing approach in subjects with a diagnosis compatible with XLRP.

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INTRODUCTION

Retinitis pigmentosa (RP), also known as rod-cone dystrophy, is a rare inherited retinal disorder with a high clinical and genetic heterogeneity (OMIM #268000). Among the different inheritance patterns of RP, the X-linked mode is responsible for one of the most severe phenotypes (X-linked RP, XLRP) affecting predominantly males. A recent study [1] estimated the prevalence of XLRP to between 4.0–5.2 per 100,000 males, and, in particular, the prevalence of XLRP due to a pathogenic variant in the retinitis pigmentosa GTPase regulator gene (*RPGR*) to between 3.4–4.4 per 100,000 males, highlighting the major contribution of this gene to the disease. The majority of males suffering from *RPGR*-XLRP present night blindness in the first decades, followed by rapid progression of peripheral and central vision loss resulting in legal blindness in the third or fourth decade [2]. Compared to males, females carrying heterozygous *RPGR* mutations usually have a more favorable visual prognosis, even though a broad spectrum of severity has been reported influenced by the random inactivation of X chromosomes ([3, 4]).

The *RPGR* gene encodes several isoforms of the RP GTPase regulator protein as a result of alternative splicing (<https://www.gtexportal.org/home/transcriptPage>). Among them, the *RPGR*^{ORF15} isoform (1,152 amino acids, NP_001030025.1), which is predominantly expressed in retina and brain ([5, 6]) plays a key role in the function of the photoreceptor cells [7]. The *RPGR*^{ORF15} transcript (NM_001034853.2) consists of the first fourteen exons of *RPGR* and a large terminal exon (exon 15 plus part of intron 15) called open reading frame 15 (ORF15), which encompasses a highly repetitive, purine-rich domain [5]. Because of this low-complexity, ORF15 is prone to mutational events and is considered as a hot spot responsible for approximately 80% of *RPGR*-XLRP cases ([5]; Leiden Open Variation Database (LOVD) Global Variome Shared Instance (LOVD GVShared, <https://databases.lovd.nl/shared/genes/RPGR>)), making its screening essential for an comprehensive molecular diagnosis. Moreover, this screening is also particularly important because ORF15 variants are usually associated with a more deleterious phenotype compared to variants located in other exons [8].

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Table 1. Results of the masked test from bioinformatic data obtained from 52 affected individuals carrying 32 different ORF15 pathogenic variants.

HGVS DNA	Genomic nomenclature	HGVS protein	Status	Bioinformatic detection		
				Run 12 8 h	Run 24 2.5 h	Run 24 2.5 x2 h
(NM_001034853.2)	(hg38)					
c.1861dup	g.38287139dup	p.(Glu621GlyfsTer9)	M	✓		
c.2153del	g.38286848del	p.(Gly718AlafsTer51)	M	✓		
c.2218G>T	g.38286781C>A	p.(Glu740Ter)	M	✓		
			F	✓		
c.2234_2237del	g.38286763_38286766del	p.(Arg745LysfsTer69)	M	✓		
			F	✓		
c.2236_2237del	g.38286765_38286766del	p.(Glu746ArgfsTer23)	M	✓		
			F	✓		
c.2257_2260del	g.38286743_38286746del	p.(Gly753LysfsTer61)	M	✓		
			F	✓		
c.2270_2271del	g.38286730_38286731del	p.(Glu757GlyfsTer12)	M	✓		
			F	✓		
c.2284G>T	g.38286715C>A	p.(Glu762Ter)	F	✓		
c.2287G>T	g.38286712C>A	p.(Glu763Ter)	F	✓		
c.2405_2406del	g.38286595_38286596del	p.(Glu802GlyfsTer32)	M	✓		
			F	✓		
c.2416del	g.38286586del	p.(Glu806ArgfsTer9)	M	✓	✓	✓
c.2426_2427del	g.38286574_38286575del	p.(Glu809GlyfsTer25)	M	✓	✓	✓
			F	x	x	✓
c.2442_2445del	g.38286554_38286557del	p.(Gly817LysfsTer2)	M	✓	✓	✓
			F	✓	✓	✓
c.2455_2468del	g.38286540_38286553del	p.(Val819ArgfsTer11)	M	✓	✓	✓
			F	✓	✓	✓
c.2468_2472del	g.38286530_38286534del	p.(Lys823ArgfsTer10)	M	✓	✓	✓
c.2506del	g.38286497del	p.(Glu836LysfsTer253)	M	✓	✓	✓
			F	x	x	x
c.2522del	g.38286477del	p.(Glu841GlyfsTer248)	M	x	x	x
c.2548G>T	g.38286451C>A	p.(Glu850Ter)	F	✓		
c.2601_2602del	g.38286399_38286400del	p.(Glu868GlyfsTer210)	M	✓		
c.2628_2629del	g.38286372_38286373del	p.(Glu877GlyfsTer201)	M	✓	✓	✓
			F	✓	✓	✓
c.2719G>T	g.38286280C>A	p.(Glu907Ter)	M	✓		
			F	✓		
c.2792del	g.38286207del	p.(Glu931GlyfsTer158)	M	x	x	x
			F	x	x	x
c.2864G>A	g.38286135C>T	p.(Trp955Ter)	M	✓		

Table 1. continued

HGVS DNA	Genomic nomenclature	HGVS protein	Status	Bioinformatic detection		
				Run 12 8 h	Run 24 2.5 h	Run 24 2.5 x2 h
(NM_001034853.2)	(hg38)		F	✓		
c.2931_2932ins [AAAGG;2908_2931]	g.38286091_38286092insCCTTTCTCTCTCCCTCCCTCCCTCTCTCC	p.(Glu978LysfsTer121)	M	×	✓	✓
c.2937_2938del	g.38286063_38286064del	p.(Glu980GlyfsTer98)	F	×	✓	×
c.2964_2965del	g.38286036_38286037del	p.(Glu989GlyfsTer89)	M	✓	✓	✓
c.2997_2998del	g.38286003_38286004del	p.(Glu1000GlyfsTer78)	M	✓	✓	✓
c.3027_3028del	g.38285973_38285974del	p.(Glu1010GlyfsTer68)	M	×	✓	✓
c.3104_3105del	g.38285897_38285898del	p.(Glu1035GlyfsTer43)	M	✓	✓	✓
c.3286_3287delinsC	g.38285712_38285713delinsG	p.(Lys1096HisfsTer35)	M	✓	✓	✓
c.3308dup	g.38285691dup	p.(Tyr1103Ter)	F	×	✓	✓
c.3308_3309del	g.106927220_106927221del	p.(Tyr1103SerfsTer7)	M	✓		

HGVS DNA (NM_001034853.2), genomic nomenclature (hg38) and HGVS protein nomenclatures are indicated for each pathogenic variant. Variations localized in the refractory region g.38,285,847 - 38,286,647 are indicated in bold. A correct bioinformatic report is indicated by the ✓ symbol and an incorrect by the × symbol.
M male, F female.

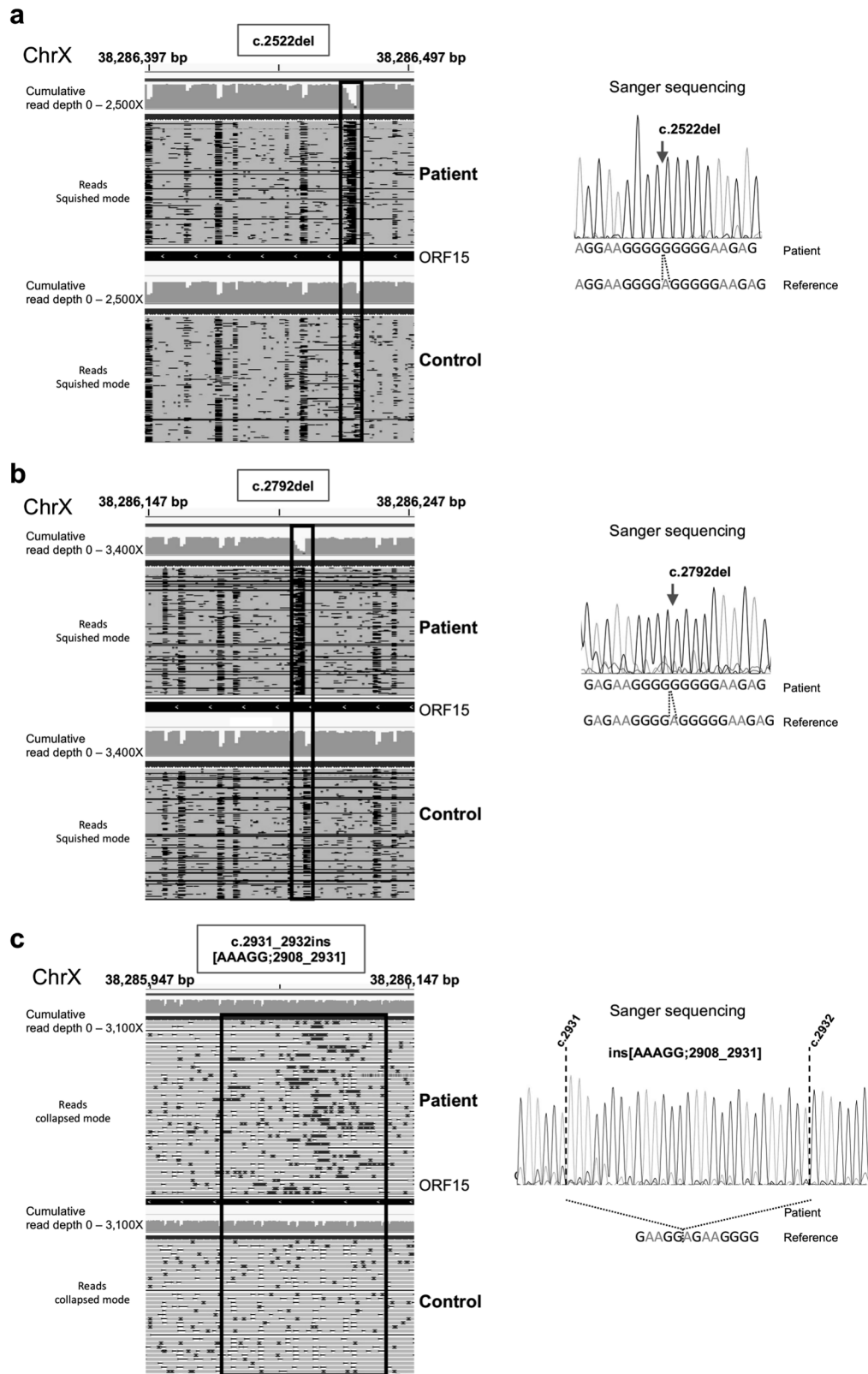


Fig. 1 Inspection of the sequenced reads using the IGV software and Sanger sequencing validation of the 3 hemizygous Clair3 undetected variants. **a** On the left: visual comparison of squished sequenced reads obtained from a male index case carrying the c.2522del and a male control. On the right: validation of the variant by Sanger sequencing. **b** On the left: visual comparison of squished sequence reads obtained from a male index case carrying the c.2792del and a male control. On the right: Validation of the variant by Sanger sequencing. **c** On the left: visual comparison of collapsed sequenced reads obtained from a male index case carrying the c.2931_2932ins[AAAGG;2908_2931] and a male control. The inserted sequences are indicated by dark gray symbols (III). On the right: Validation of the variant by Sanger sequencing. The reads show a representative subset in each IGV capture. The visual difference observed between a patient and its control is highlighted by a black frame. An arrow in Sanger electropherograms indicates the genomic position of each pathogenic variant.

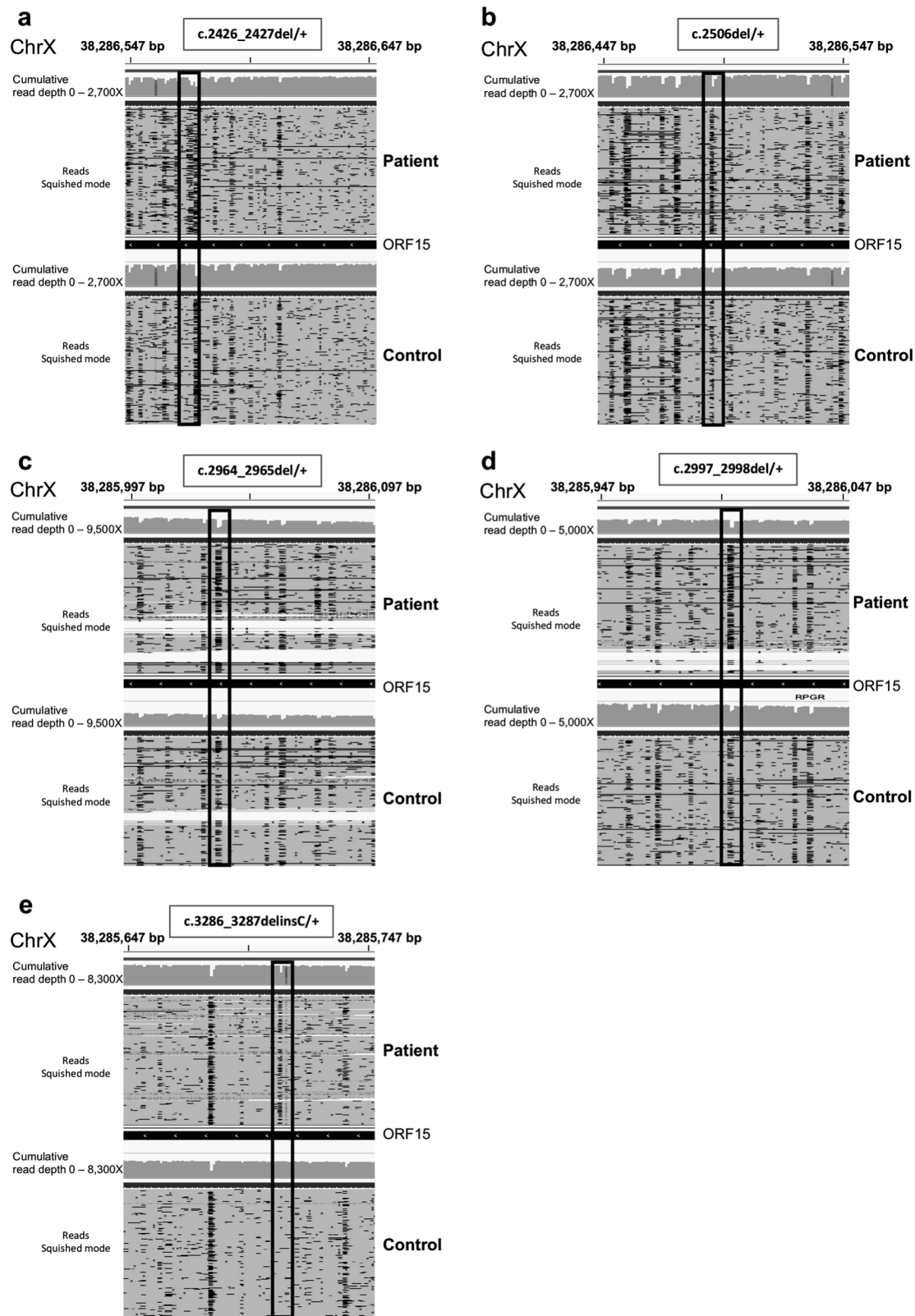


Fig. 2 Inspection of the sequenced reads using the IGV software and Sanger sequencing validation of 5 heterozygous *Clair3* undetected variants. For each studied pathogenic variant, a visual comparison of squished sequence reads obtained from a carrier female and a female control is shown. The heterozygous variants are: **a** c.2426_2427del, **b** c.2506del, **c** c.2964_2965del, **d** c.2997_2998del and **e** c.3286_3287delinsC. The reads show a representative subset in each IGV capture. The visual difference observed between a patient and its control is highlighted by a black frame.